# NEURAL NETWORK PRACTICAL PROGRAMS

*Experiments 1–8*

## Experiment 1: Create a Simple Artificial Neural Network Using Python

```
import numpy as np

x=np.array([2,3])
w=np.array([0.5,0.5])
b=1
y=np.dot(x,w)+b
print(y)
```

## Experiment 2: Plot Sigmoid and ReLU Activation Functions

```
import numpy as np
import matplotlib.pyplot as plt

x=np.linspace(-5,5,100)
sig=1/(1+np.exp(-x))
relu=np.maximum(0,x)

plt.plot(x,sig,label="Sigmoid")
plt.plot(x,relu,label="ReLU")
plt.legend()
plt.show()
```

## Experiment 3: Implement a Feedforward Neural Network

```
import numpy as np

x=np.array([2,3])
w=np.array([0.5,0.5])
b=1
h=np.maximum(0,np.dot(x,w)+b)
print(h)
```

## Experiment 4: Train a Simple Neural Network Classifier

```
import numpy as np

x=np.array([[1,2],[2,3],[3,4]])
y=np.array([0,0,1])
w=np.array([1,1])
```

```
p=(x@w>4).astype(int)
print(p)
```

## Experiment 5: Perform Image Convolution Operations

```
import tensorflow as tf
import numpy as np

img=np.random.rand(1,5,5,1).astype("float32")
k=np.ones((3,3,1,1))

out=tf.nn.conv2d(img,k,1,"VALID")
print(out.numpy())
```

## Experiment 6: Apply Image Filters — Edge Detection and Blur

```
import tensorflow as tf
import numpy as np

img=np.random.rand(1,5,5,1).astype("float32")

edge=tf.image.sobel_edges(img)
blur=tf.nn.avg_pool2d(img,3,1,"SAME")

print(edge.shape)
print(blur.numpy())
```

## Experiment 7: Visualize CNN Feature Maps

```
import tensorflow as tf
import matplotlib.pyplot as plt
import numpy as np

img=np.random.rand(1,8,8,1).astype("float32")
k=np.random.rand(3,3,1,2).astype("float32")

f=tf.nn.conv2d(img,k,1,"SAME")

plt.imshow(f[0,:,:,0],cmap="gray")
plt.show()
```

## Experiment 8: Implement Max Pooling and Average Pooling

```
import tensorflow as tf
import numpy as np

x=np.array([[[[1],[2]],[[3],[4]]]],dtype="float32")

print(tf.nn.max_pool2d(x,2,2).numpy())
print(tf.nn.avg_pool2d(x,2,2).numpy())
```

## Run This Command in Terminal

Copy and paste this command into the terminal:

```
pip install numpy matplotlib tensorflow
```

## Required Libraries

| Library | Purpose |
|---|---|
| NumPy | Numerical operations |
| Matplotlib | Graphs and visualization |
| TensorFlow | CNN and pooling operations |